3.6 FSE / TSE & EPI SEQUENCES

FAST WHEN YOU NEED IT. ULTRA-FAST WHEN EVERY SECOND COUNTS.

FSE (Fast Spin Echo / TSE – Turbo Spin Echo) acquires multiple echoes per TR using a 180° refocusing train.
EPI (Echo Planar Imaging) fills k-space after a single excitation with a rapid echo train.

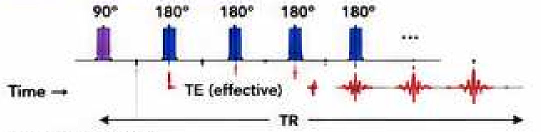
KEY CONCEPT

- FSE/TSE = multiple echoes per TR → faster, SNR efficient, less sensitive to motion.
- EPI = single excitation, rapid readout → fastest imaging, most sensitive to artifacts.
- Pick the right tool for the clinical goal and patient condition.
- Understand trade-offs: speed, SNR, resolution, and artifacts.

↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW THEY WORK

A. FSE / TSE (Turbo Spin Echo)



- One 90° excitation.
- A train of 180° refocusing pulses produces multiple spin echoes.
- Each echo fills a different line of k-space.
- Result: faster imaging with high SNR and less motion sensitivity.

A.K.A. FSE, TSE, RARE, FRFSE

B. EPI (Echo Planar Imaging)

- Single 90° excitation.
- Entire k-space (lines) acquired after that excitation with a rapid echo train.
- Result: extremely fast (single shot) but more susceptible to artifacts.

A.K.A. SE-EPI, GRE-EPI, EPI-DWI, fMRI EPI

2 KEY DIFFERENCES: FSE/TSE vs EPI

Feature	FSE / TSE	EPI
Excitations	Multiple echoes per TR	Single excitation
Speed	Fast	Ultra-fast (single shot)
SNR	High (many echoes)	Lower (one excitation)
Motion Sensitivity	Low / Moderate	High
Susceptibility Sensitivity	Low / Moderate	High
Distortion	Low	High
Long TE Capability	Yes (very long TE)	Limited (shorter TE)
Spatial Resolution	High	Lower (typically)
Bandwidth	Moderate	High (needed)
Best For	Anatomic detail	Speed critical / Functional
Examples	T2 FSE, PD FSE, STIR FSE	EPI DWI, fMRI, PWI

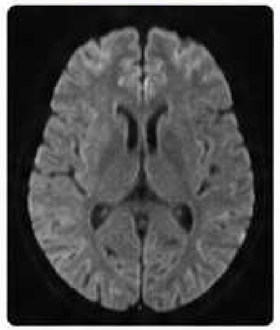
Rule of thumb: If you can wait a little → use FSE/TSE for better quality. If you can't wait → use EPI for speed.

3 IMAGE APPEARANCE EXAMPLES (1.5T)

FSE / TSE (T2-weighted)

- Excellent T2 contrast
- High SNR
- Less distortion
- Good anatomical detail

EPI (DWI b=1000)



- Very fast (single shot)
- Susceptibility distortion
- Lower SNR
- Detects restricted diffusion

★ Expect EPI to look "noisier" and more distorted, especially near air/tissue or metal.

4 WHEN TO USE WHICH?

Clinical Goal / Scenario	Prefer	Why?
Routine anatomy (brain, spine, joints)	FSE / TSE	Better SNR, resolution, contrast, less artifacts.
Long T2 contrast needed (edema, fluid)	FSE / TSE	Very long TE possible with high quality.
STIR Imaging	FSE / TSE	Uniform fat suppression, high SNR.
Diffusion imaging (DWI)	EPI	Single shot, diffusion gradients require speed.
fMRI (BOLD)	EPI	Repeated whole-brain volumes every few seconds.
Perfusion (PWI)	EPI	Fast dynamic imaging essential.
Breath-hold abdomen	EPI	Capture data in 1–2 seconds.
Uncooperative / moving patient	EPI or FSE	EPI if motion too much; FSE if quality needed.
Low field (0.3T)	FSE / TSE	Better SNR efficiency.

5 COMMON FSE / TSE VARIANTS

Variant	Key Feature	Typical Use
T2 FSE	Long TE, high echo train	General T2 anatomy
PD FSE	Short TE, long TR	Soft tissue detail
STIR FSE	Inversion recovery + FSE	Edema, marrow, fat suppr.
3D FSE (SPACE / CUBE)	3D volume with FSE readout	Neuro, joints, high res
Double Echo FSE	Two echoes per TR	Cartilage, joints

★ Longer echo train length (ETL) = faster scan but more blurring and signal decay.

6 EPI VARIANTS & COMMON USES

Variant	Key Feature	Typical Use
SE-EPI (DWI)	Spin echo EPI with diffusion grads	Stroke, tumors, acute Dx
GRE-EPI (fMRI)	Gradient echo EPI	BOLD fMRI
EPI-PWI	Dynamic GRE-EPI	Perfusion (brain, body)
ASSET / SENSE EPI	Parallel imaging + EPI	Faster, less distortion
Multi-shot EPI	Breaks k-space into shots	Less distortion, longer time

★ Higher receiver bandwidth in EPI reduces distortion but lowers SNR.

7 ARTIFACTS: FSE/TSE vs EPI

FSE / TSE Artifacts (Less Common)

- T2 blurring (long ETL)
- Echo train modulation (ETL too long)
- Gibbs (truncation)
- Motion (ghosting if severe)
- Chemical shift (frequency direction)

EPI Artifacts (More Common)

- Geometric distortion (B_0 inhomogeneity)
- Susceptibility (air/tissue, metal)
- Nyquist ghosting (odd/even echoes)
- Motion ghosting / blur
- EPI dropout (signal loss)
- Eddy currents (DWI)

For EPI: improve shimming, raise bandwidth, short TE, parallel imaging, and minimize motion.

8 ENGINEER CHECKLIST

- ☑ Correct sequence selected for clinical goal?
- ☑ FSE: Echo train length optimized for SNR vs blurring?
- ☑ EPI: Phase encode direction chosen to minimize distortion?
- ☑ Prescan / Center Frequency done at correct location?
- ☑ Shim quality good? (EPI very sensitive)
- ☑ Bandwidth high enough for EPI?
- ☑ Parallel imaging enabled when appropriate?
- ☑ Coil elements functioning and properly connected?
- ☑ No gradient faults or RF transmit/receive errors?
- ☑ Repeat test or sequence to confirm if needed.

QUICK TIP

If EPI looks distorted or blurry: think SHIM → BANDWIDTH → PHASE DIR → MOTION.

9 CLINICAL EXAMPLES

Acute Stroke	DWI (EPI)	Detect restricted diffusion within minutes.
Spine Anatomy	T2 FSE	Best CSF detail, disc and nerve roots.
Knee / MSK	PD FSE + STIR FSE	Edema, marrow, ligaments.
Cardiac fMRI	GRE-EPI	BOLD functional imaging.
Liver / Abdomen	T2 FSE	High quality anatomy, lesion detection.

10 PARAMETERS THAT MATTER MOST

FSE / TSE		EPI	
Echo Train Length (ETL)	Speed ↑, blurring ↑	TE (Very Short)	Shorter → less distortion, ↑ SNR
TR	↑ SNR & T1 recovery	Bandwidth (High)	Reduces distortion & ghosting
TE	Contrast (T2 ↑ with longer TE)	Phase Encode Dir.	A→P (brain) or R→L (abdomen)
Echo Spacing	Short spacing → less blurring	Parallel Imaging	Less distortion, faster
Bandwidth	↑ BW → less blurring, ↓ SNR	Echo Spacing	Short spacing → less blurring

Optimize parameters for the patient, anatomy, and clinical question.

11 PRACTICAL TIPS

- Use FSE/TSE for highest image quality.
- Use EPI when speed is critical.
- Check shim on EPI before scanning.
- Higher bandwidth is your friend in EPI.
- Minimize motion—use padding, coaching, or gating.
- Review first images—fix early, not late.
- Compare with other sequences for confidence.

REMEMBER

FSE/TSE = fast with quality.
EPI = ultra-fast with trade-offs.
The right sequence at the right time delivers the right answer.

KEY TAKEAWAY

Understand the strengths and weaknesses of FSE/TSE and EPI. Your choices impact speed, image quality, diagnosis, and patient care.

★ **RIGHT INFORMATION. RIGHT BALANCE. RIGHT IMAGE.** Understand the sequence. Recognize the trade-offs. Support the technologist. Deliver the value.